A Bounded-Hessian-Lipschitz Counterexample to the Dimension-Free Logarithmic Local Rate

July 14, 2026

Abstract

We give an option-(B) counterexample to the Hessian-Lipschitz local-rate conjecture. A two-scale radial-softplus construction produces smooth genuine Hessians, already in equilibrium-whitened coordinates, for which

$$\kappa \asymp \frac{d}{\sigma^2}, \qquad \gamma_{\mathrm{loc}} \asymp \frac{\sigma}{\sqrt{d}} \asymp \kappa^{-1/2}, \qquad L_H \asymp \frac{\sqrt{d}}{\sigma^2}, \qquad \tau_H^2 \gtrsim \frac{\sqrt{d}}{\sigma} \asymp \sqrt{\kappa}.$$

Taking $(d, \sigma) = (n^4, n)$ gives $L_H \leq \Lambda_0$ for one absolute constant, $\kappa \asymp n^2$, and $\gamma_{\mathrm{loc}} \asymp n^{-1} \asymp \kappa^{-1/2}$. Thus bounded whitened Hessian-Lipschitz modulus does not rescue a logarithmic local rate. The proof uses exact Gaussian isotropization and a literally Fisher–Rao-normalized affine Rayleigh witness. No Frobenius/operator conversion, symmetry-only surrogate, or noncommutative Loewner transfer occurs.

1 Main result and norm convention

For a C^3 potential define its Hessian operator-Lipschitz modulus by

$$L_H(V) := \sup_{x \neq y} \frac{\|\nabla^2 V(x) - \nabla^2 V(y)\|_{\mathrm{op}}}{\|x - y\|}.$$

Since the third derivative is symmetric,

$$L_H(V) = \sup_x \sup_{\|h\|=\|u\|=1} |\nabla^3 V(x)[h, u, u]|. \tag{1.1}$$

Indeed, the right side is $\sup_{x, \|h\|=1} \|D(\nabla^2 V)(x)[h]\|_{\mathrm{op}}$; the fundamental theorem of calculus proves one inequality and directional difference quotients prove the other. Thus every L_H below is the tensor/operator quantity in (H2), not a Frobenius norm. For a symmetric trilinear form $\mathcal{T}$, we also write $\|\mathcal{T}\|_{\mathrm{inj}} := \sup_{\|h\|=\|u\|=\|v\|=1} |\mathcal{T}[h, u, v]|$.

Theorem 1.1 (Bounded-L_H square-root counterexample). *There are numerical constants $c, C > 0$ and $n_0 \in \mathbb{N}$ such that for every integer $n \geq n_0$ there is a C^∞, strongly convex potential $V_n : \mathbb{R}^{n^4+1} \to \mathbb{R}$ satisfying, for $Z \sim \mathcal{N}(0, I_{n^4+1})$,*

$$\mathbb{E}\nabla V_n(Z) = 0, \qquad \mathbb{E}\nabla^2 V_n(Z) = I. \tag{1.2}$$

If α_n, β_n are its optimal scalar curvature constants and $\kappa_n = \beta_n/\alpha_n$, then

$$\frac{c}{n} \leq \alpha_n \leq \frac{C}{n}, \qquad cn \leq \beta_n \leq Cn, \qquad cn^2 \leq \kappa_n \leq Cn^2. \tag{1.3}$$

Its whitened Hessian-Lipschitz modulus and linearized gap obey

$$c \le L_H(V_n) \le C, \qquad \frac{c}{n} \le \gamma_{\mathrm{loc}}(V_n) \le \frac{C}{n}, \tag{1.4}$$

and its first-Hermite Hessian coupling satisfies

$$\tau_H(V_n)^2 \ge cn. \tag{1.5}$$

Consequently

$$\gamma_{\mathrm{loc}}(V_n) \asymp \kappa_n^{-1/2}, \qquad \gamma_{\mathrm{loc}}(V_n)^{-1} \asymp \sqrt{\kappa_n} \gg \log \kappa_n.$$

In particular, for an absolute Λ_0 the whole sequence belongs to $\mathcal{C}(\kappa_n, \Lambda_0)$, while $\gamma_{\mathrm{loc}}(V_n)(1+\log\kappa_n) \to 0$.

The construction and proof occupy Sections 2–6.

2 The two-scale radial-softplus family

Let d and σ satisfy

$$\sigma \ge 1, \qquad d \ge C_0\sigma^2, \tag{2.1}$$

where C_0 is a sufficiently large numerical constant. Write $z = (t, y) \in \mathbb{R} \times \mathbb{R}^d$ and set

$$r(y) := \sqrt{\sigma^2 + \|y\|^2}, \qquad R := \sqrt{\sigma^2 + d}, \qquad F_\sigma(s) := \sigma \log(1 + e^{s/\sigma}). \tag{2.2}$$

Put

$$p(s) := F'_\sigma(s), \qquad q(s) := F''_\sigma(s) = \frac{1}{\sigma} p(s)(1 - p(s)).$$

Thus $0 < p < 1$, $0 < q \le (4\sigma)^{-1}$, and $|F'''_\sigma| \le (4\sigma^2)^{-1}$.

For $0 \le \delta \le 1$, define

$$s_\delta(t, y) := r(y) - R - \delta t, \qquad U_\delta(t, y) := F_\sigma(s_\delta(t, y)), \qquad A_\delta := \nabla^2 U_\delta. \tag{2.3}$$

Writing $S = \|y\|^2$,

$$v_\delta := \nabla s_\delta = (-\delta, y/r), \qquad D(y) := \nabla_y^2 r = \frac{I_d}{r} - \frac{yy^\top}{r^3}. \tag{2.4}$$

The exact Hessian is

$$A_\delta = q(s_\delta) v_\delta v_\delta^\top + p(s_\delta) \begin{pmatrix} 0 & 0 \\ 0 & D(y) \end{pmatrix}. \tag{2.5}$$

Both terms are positive semidefinite. The tangential eigenvalue of D is r^{-1} and its radial eigenvalue is $\sigma^2 r^{-3}$. Since $\|v_\delta\|^2 \le 2$,

$$0 \preceq A_\delta \preceq \frac{3}{2\sigma} I_{d+1}. \tag{2.6}$$

3 Exact Gaussian isotropization

Let $T \sim \mathcal{N}(0,1)$ and $Y \sim \mathcal{N}(0, I_d)$ be independent. In the following expectations $s_\delta = s_\delta(T,Y)$ and $r = r(Y)$. Define

$$Q_{d,\sigma}(\delta) := \mathbb{E} q(s_\delta), \tag{3.1}$$

$$M_{d,\sigma}(\delta) := \frac{1}{d}\mathbb{E}\left[q(s_\delta)\frac{S}{r^2} + p(s_\delta)\left(\frac{d-1}{r} + \frac{\sigma^2}{r^3}\right)\right]. \tag{3.2}$$

Rotation invariance and oddness of the mixed block give

$$\mathbb{E}A_\delta = \operatorname{diag}(\delta^2 Q_{d,\sigma}(\delta), M_{d,\sigma}(\delta) I_d). \tag{3.3}$$

Lemma 3.1 (Uniform scalar estimates and the isotropizing root)**.** *Under* (2.1)*, uniformly for* $0 \le \delta \le 1$*,*

$$\frac{c}{\sigma} \le Q_{d,\sigma}(\delta) \le \frac{1}{4\sigma}, \qquad \frac{c}{R} \le M_{d,\sigma}(\delta) \le \frac{C}{R}. \tag{3.4}$$

If C_0 is large enough, the continuous equation

$$\delta^2 Q_{d,\sigma}(\delta) = M_{d,\sigma}(\delta) \tag{3.5}$$

has a root $\delta_{d,\sigma} \in (0,1)$. For any such root, with $m_{d,\sigma}$ denoting the common value,

$$\delta_{d,\sigma}^2 \asymp \frac{\sigma}{R}, \qquad m_{d,\sigma} \asymp \frac{1}{R}, \qquad \mathbb{E}A_{\delta_{d,\sigma}} = m_{d,\sigma} I_{d+1}. \tag{3.6}$$

Proof. On the event

$$\mathcal{E}_d := \{|S - d| \le 2\sqrt{d},\ |T| \le 1\},$$

Chebyshev's inequality and independence give $\mathbb{P}(\mathcal{E}_d) \ge c_0 > 0$. Moreover

$$|r - R| = \frac{|S-d|}{r+R} \le 2, \qquad |s_\delta| \le 3.$$

Since $\sigma \ge 1$, both $p(s_\delta)$ and $\sigma q(s_\delta)$ are bounded below on this event. This proves the lower bound for Q. On the same event,

$$\operatorname{Tr} D = \frac{d-1}{r} + \frac{\sigma^2}{r^3} \ge c\frac{d}{R},$$

which proves the lower bound for M.

The upper bound for Q is immediate. For $d > 2$,

$$\mathbb{E}\frac{1}{r} \le \mathbb{E}S^{-1/2} \le (\mathbb{E}S^{-1})^{1/2} = \frac{1}{\sqrt{d-2}} \le \frac{C}{R}. \tag{3.7}$$

The first summand in (3.2) is at most $(4\sigma d)^{-1}$, and the second is at most $(1 + d^{-1})\mathbb{E}r^{-1}$. This proves (3.4).

Continuity follows by dominated convergence. The left side minus the right side of (3.5) is negative at zero and positive at one once R/σ is a sufficiently large numerical constant. Hence a root exists. At any root, (3.4) gives (3.6), and (3.3) gives exact isotropy. $\square$

Fix a root, abbreviate $U = U_{\delta_{d,\sigma}}$, $A = \nabla^2 U$, and set

$$\alpha := \frac{\sigma}{R}, \qquad c_{d,\sigma} := \frac{1-\alpha}{m_{d,\sigma}} \asymp R. \tag{3.8}$$

Define

$$V_{d,\sigma}(z) := \frac{\alpha}{2}\|z\|^2 + c_{d,\sigma}U(z) + b_{d,\sigma}^\top z, \qquad b_{d,\sigma} := -c_{d,\sigma}\mathbb{E}\nabla U(Z). \tag{3.9}$$

The gradient of U is bounded, so $b_{d,\sigma}$ is finite. It is a multiple of the t direction. The Hessian

$$W_{d,\sigma} = \nabla^2 V_{d,\sigma} = \alpha I + c_{d,\sigma}A \tag{3.10}$$

satisfies, exactly,

$$\mathbb{E}\nabla V_{d,\sigma}(Z) = 0, \qquad \mathbb{E}W_{d,\sigma}(Z) = I. \tag{3.11}$$

The optimal lower curvature is α, since $A(t, y) \to 0$ as $t \to +\infty$ for fixed y. Equation (2.6) gives the upper curvature bound. Conversely, at $y = 0$ and with t chosen so that $s_\delta = 0$, the transverse block of A is $I_d/(2\sigma)$. Thus the optimal upper curvature β satisfies

$$\alpha + \frac{c_{d,\sigma}}{2\sigma} \le \beta \le \alpha + \frac{3c_{d,\sigma}}{2\sigma}, \qquad \beta \asymp \frac{R}{\sigma}, \qquad \kappa := \frac{\beta}{\alpha} \asymp \frac{R^2}{\sigma^2} \asymp \frac{d}{\sigma^2}. \tag{3.12}$$

The potential is C^∞ and α-strongly convex, hence normalizable. It is a genuine Hessian and its complete compatibility hierarchy holds classically.

4 The exact Hessian-Lipschitz scale

The third derivative of r is

$$\nabla^3 r[h, u, v] = -\frac{(y\cdot h)(u\cdot v) + (y\cdot u)(h\cdot v) + (y\cdot v)(h\cdot u)}{r^3} \tag{4.1}$$

$$+ 3\frac{(y\cdot h)(y\cdot u)(y\cdot v)}{r^5}. \tag{4.2}$$

Consequently

$$\|\nabla s_\delta\| \le \sqrt{2}, \qquad \|\nabla^2 s_\delta\|_{\mathrm{op}} \le \sigma^{-1}, \qquad \|\nabla^3 s_\delta\|_{\mathrm{inj}} \le 6\sigma^{-2}. \tag{4.3}$$

The exact chain rule is

$$\nabla^3 U[h, u, v] = F_\sigma'''(s)\, ds[h]ds[u]ds[v] \tag{4.4}$$

$$+ q(s)\{\nabla^2 s[h, u]ds[v] + \nabla^2 s[h, v]ds[u] + \nabla^2 s[u, v]ds[h]\} \tag{4.5}$$

$$+ p(s)\nabla^3 s[h, u, v]. \tag{4.6}$$

Equations (4.3)–(4.4) give

$$\sup_z \|\nabla^3 U(z)\|_{\mathrm{inj}} \le \frac{6 + 5\sqrt{2}/4}{\sigma^2} < \frac{8}{\sigma^2}. \tag{4.7}$$

For the reverse bound, fix $y = (\sigma/2)e_1$ and send $t \to -\infty$. Then $p(s) \to 1$ and $q(s), F_\sigma'''(s) \to 0$, so $\nabla^3 U \to \nabla^3 r$. Directly from (4.1),

$$|\nabla^3 r[e_1, e_1, e_1]| = \frac{c_1}{\sigma^2}$$

for a numerical $c_1 > 0$. Since the quadratic and affine terms in (3.9) have zero third derivative,

$$L_H(V_{d,\sigma}) \asymp \frac{c_{d,\sigma}}{\sigma^2} \asymp \frac{R}{\sigma^2} \asymp \frac{\sqrt{d}}{\sigma^2}. \tag{4.8}$$

This is a direct injective-tensor estimate; no dimension-dependent Frobenius conversion is present.

5 Compatible Rayleigh identity and the inverse gap

If X is the physical covariance perturbation, its packed coordinate is $X/\sqrt{2}$ and its Fisher norm is

$$\|(u, X)\|_\star^2 := \|u\|^2 + \frac{1}{2}\|X\|_F^2. \tag{5.1}$$

Lemma 5.1 (Compatible Bochner–Rayleigh identity)**.** *Let $V \in C^4$ and suppose that its Hessian $W = \nabla^2 V$ and the first two derivatives of W have at most polynomial growth. If $\mathbb{E}W = I$, then the packed generator in the question satisfies*

$$4\mathcal{Q}_W(u, X) = \mathbb{E}[(2u + XZ)^\top W(Z)(2u + XZ)] + \|X\|_F^2. \tag{5.2}$$

If $W \succeq \alpha I$ and $0 < \alpha \le 1$, then

$$\mathcal{Q}_W(u, X) \ge \alpha\|(u, X)\|_\star^2, \qquad \gamma_{\text{loc}}(W) \ge \alpha. \tag{5.3}$$

Proof. Expanding the displayed block operator in the packed coordinate gives

$$\mathcal{Q}_W(u, X) = \|u\|^2 + u^\top T[X] + \frac{1}{2}\|X\|_F^2 + \frac{1}{4}\langle X, H[X]\rangle_F.$$

Gaussian integration by parts and third-derivative symmetry give

$$\mathbb{E}[u^\top WXZ] = u^\top T[X].$$

In components, double Gaussian integration by parts gives

$$\mathbb{E}[Z_a Z_b W_{ij}(Z)] = \delta_{ab}\delta_{ij} + \mathbb{E}[\partial_{ab} W_{ij}(Z)].$$

Full fourth-derivative symmetry then gives

$$X_{ia}X_{jb}\mathbb{E}[\partial_{ab}W_{ij}] = X_{ij}X_{ab}\mathbb{E}[\partial_{ab}W_{ij}],$$

and therefore

$$\mathbb{E}[(XZ)^\top W(XZ)] = \|X\|_F^2 + \langle X, H[X]\rangle_F.$$

Substitution proves (5.2). The displayed family has a bounded Hessian and bounded derivatives of the Hessian of every order because $r \ge \sigma > 0$, so all steps are classical. Finally,

$$4\mathcal{Q}_W(u, X) \ge 4\alpha\|u\|^2 + (1 + \alpha)\|X\|_F^2 \ge 4\alpha\|(u, X)\|_\star^2,$$

which proves (5.3). □

For the matching upper bound, put $P_d := \operatorname{diag}(0, I_d)$ and first take the unnormalized pair

$$u_0 = e_t, \qquad X_0 = \lambda P_d, \qquad \lambda := \frac{2\delta_{d,\sigma}R}{d}. \tag{5.4}$$

Its literal Fisher–Rao normalization is

$$N_F^2 := 1 + \frac{d\lambda^2}{2}, \qquad \widehat{u} := \frac{u_0}{N_F}, \qquad \widehat{X} := \frac{X_0}{N_F}, \qquad \|(\widehat{u}, \widehat{X})\|_\star = 1. \tag{5.5}$$

The covariance coordinate in the packed Euclidean display of $L_\star$ is $\widehat{X}/\sqrt{2}$. For the unnormalized pair, $2u_0 + X_0 Z = (2, \lambda Y)$ and

$$\|X_0\|_F^2 = d\lambda^2 = \frac{4\delta_{d,\sigma}^2 R^2}{d} \le C\frac{\sigma}{R} = C\alpha, \qquad N_F^2 = 1 + O(\alpha). \tag{5.6}$$

Let $g(S) = S/r$. From (2.5), exactly,

$$(2, \lambda Y)^\top A(2, \lambda Y) = q(s)\lambda^2 \left(g(S) - \frac{d}{R}\right)^2 + p(s)\lambda^2 \frac{\sigma^2 S}{r^3}. \tag{5.7}$$

The factorization

$$g(S) - \frac{d}{R} = (S - d)\frac{\sigma^2 + rR}{rR(r + R)} \tag{5.8}$$

implies

$$\left|g(S) - \frac{d}{R}\right| \le \frac{|S - d|}{R}, \qquad \mathbb{E}\left(g(S) - \frac{d}{R}\right)^2 \le \frac{2d}{R^2} \le 2. \tag{5.9}$$

Also, by (3.7),

$$\mathbb{E}\frac{S}{r^3} \le \mathbb{E}\frac{1}{r} \le \frac{C}{R}. \tag{5.10}$$

Using $q \le (4\sigma)^{-1}$, $\delta_{d,\sigma}^2 \asymp \sigma/R$, and $c_{d,\sigma} \asymp R$, equations (5.7)–(5.10) give

$$c_{d,\sigma}\mathbb{E}[(2, \lambda Y)^\top A(2, \lambda Y)] \le C\left(\frac{1}{d} + \frac{\sigma^3 R}{d^2}\right) \le C\frac{\sigma}{R} = C\alpha. \tag{5.11}$$

Moreover

$$\alpha\mathbb{E}\|(2, \lambda Y)\|^2 = \alpha(4 + d\lambda^2) \le C\alpha. \tag{5.12}$$

Substitution into (5.2) proves $\mathcal{Q}_W(u_0, X_0) \le C\alpha$. By homogeneity and (5.5),

$$\gamma_{\mathrm{loc}}(V_{d,\sigma}) \le \mathcal{Q}_W(\widehat{u}, \widehat{X}) = \frac{\mathcal{Q}_W(u_0, X_0)}{N_F^2} \le C\frac{\sigma}{R}. \tag{5.13}$$

The alpha floor gives the reverse bound. Together with (3.12),

$$\gamma_{\mathrm{loc}}(V_{d,\sigma}) \asymp \frac{\sigma}{R} \asymp \kappa^{-1/2}. \tag{5.14}$$

5.1 Symmetry-sector check

The construction is $O(d)$-invariant in the y variables. Accordingly the packed mean–covariance space splits orthogonally into a three-dimensional trivial sector

$$\mathrm{span}\{e_t;\ e_t e_t^\top;\ P_d/\sqrt{d}\},$$

a two-dimensional standard sector $\mathrm{span}\{e_j; (e_t e_j^\top + e_j e_t^\top)/\sqrt{2}\}$ repeated d times, and a transverse-traceless scalar sector of multiplicity $d(d+1)/2 - 1$. The multiplicities sum to $(d+1) + (d+1)(d+2)/2$, the dimension of the full packed space.

For an exact check, write

$$W = \begin{pmatrix} a & by^\top \\ by & c_0 I_d + e\, yy^\top \end{pmatrix}$$

with

$$a = \alpha + c_{d,\sigma}\delta^2 q, \quad b = -c_{d,\sigma}\delta q/r, \quad c_0 = \alpha + c_{d,\sigma}p/r, \quad e = c_{d,\sigma}(q/r^2 - p/r^3).$$

Let $E = e_t e_t^\top$, $P = P_d/\sqrt{d}$, and set

$$t_E = \mathbb{E}[Ta], \qquad t_P = \frac{1}{\sqrt{d}}\mathbb{E}[T(dc_0 + eS)] = -\frac{c_{d,\sigma}\delta}{\sqrt{d}}\mathbb{E}[qS/r],$$

$$h_{EE} = \mathbb{E}[a(T^2 - 1)], \qquad h_{EP} = \mathbb{E}[a(S-d)/\sqrt{d}] = \frac{1}{\sqrt{d}}\mathbb{E}[(dc_0 + eS)(T^2 - 1)],$$

$$h_{PP} = \mathbb{E}[(dc_0 + eS)(S-d)/d].$$

In the orthonormal packed basis (e_t, E, P), the trivial block is exactly

$$L_{\text{triv}} = \begin{pmatrix} 1 & t_E/\sqrt{2} & t_P/\sqrt{2} \\ t_E/\sqrt{2} & 1 + h_{EE}/2 & h_{EP}/2 \\ t_P/\sqrt{2} & h_{EP}/2 & 1 + h_{PP}/2 \end{pmatrix}. \tag{5.15}$$

The normalized witness (5.5) has the exact sector coordinate

$$\frac{(1, 0, \lambda\sqrt{d/2})}{\sqrt{1 + d\lambda^2/2}} \tag{5.16}$$

in (5.15). Its block Rayleigh quotient is therefore identically the full-space quotient evaluated in (5.7)–(5.13); no multiplicity or physical-versus-packed coordinate is being suppressed.

6 First-Hermite certificate

Let

$$B := \operatorname{diag}(0, I_d/\sqrt{d}), \qquad \|B\|_F = 1. \tag{6.1}$$

Hessian compatibility and two Gaussian integrations by parts give

$$\begin{aligned} e_t^\top T[B] &= \mathbb{E}[t\,\mathrm{Tr}(BW_{d,\sigma})] = \mathbb{E}[(BZ)^\top W_{d,\sigma} e_t] \\ &= -\frac{c_{d,\sigma}\delta_{d,\sigma}}{\sqrt{d}}\mathbb{E}\left[q(s)\frac{S}{r}\right]. \end{aligned} \tag{6.2}$$

On the event $\mathcal{E}_d$ from Lemma 3.1, $q(s) \ge c/\sigma$ and $S/r \ge c\sqrt{d}$. Hence

$$\tau_H^2 \ge |e_t^\top T[B]|^2 \ge c\frac{c_{d,\sigma}^2 \delta_{d,\sigma}^2}{\sigma^2} \ge c\frac{R}{\sigma} \asymp \sqrt{\kappa}. \tag{6.3}$$

This explicitly realizes the central Frobenius trap rather than using it: the averaged t-derivative of the transverse Hessian has tiny individual eigenvalues but rank d, so its Frobenius norm is large even though the pointwise injective third-tensor norm is $L_H \asymp R/\sigma^2$.

Combining (3.12), (4.8), (5.14), and (6.3) yields the complete two-parameter frontier

$$\boxed{\kappa \asymp \frac{d}{\sigma^2}, \qquad L_H \asymp \frac{\sqrt{d}}{\sigma^2}, \qquad \gamma_{\text{loc}} \asymp \frac{\sigma}{\sqrt{d}} \asymp \kappa^{-1/2}, \qquad \tau_H^2 \gtrsim \frac{\sqrt{d}}{\sigma} \asymp \sqrt{\kappa}.} \tag{6.4}$$

The specialization $(d, \sigma) = (n^4, n)$ proves Theorem 1.1. The alternative $(d, \sigma) = (n^6, n^2)$ even has $L_H \asymp n^{-1} \to 0$ with the same $\kappa \asymp n^2$ and gap scale. More generally, keeping $d \asymp \kappa\sigma^2$ leaves the square-root gap unchanged while $L_H \asymp \sqrt{\kappa}/\sigma$ can be made arbitrarily small by increasing the dimension.

7 An explicit option-(B) sequence

We record loose numerical constants so that the class quantifiers do not rest on asymptotic notation. For every integer $n \geq 15000$, set

$$d_n = n^4, \qquad \sigma_n = n, \qquad R_n = \sqrt{d_n + \sigma_n^2}, \qquad N_\theta^{(n)} = d_n + 1. \tag{7.1}$$

Choose the smallest root $\delta_n \in (0, 1)$ of (3.5), put

$$m_n := \delta_n^2 Q_{d_n,\sigma_n}(\delta_n) = M_{d_n,\sigma_n}(\delta_n), \quad \alpha_n := \frac{\sigma_n}{R_n}, \quad c_n := \frac{1-\alpha_n}{m_n},$$

and define V_n by (3.9), with these specialized parameters. Also write

$$A_n := A_{\delta_n}, \qquad W_n := \nabla^2 V_n, \qquad \lambda_n := \frac{2\delta_n R_n}{d_n}, \qquad N_{F,n}^2 := 1 + \frac{d_n \lambda_n^2}{2}, \tag{7.2}$$

and set $\widehat{u}_n = e_t/N_{F,n}$, $\widehat{X}_n = \lambda_n P_{d_n}/N_{F,n}$. Then $\|(\widehat{u}_n, \widehat{X}_n)\|_\star = 1$ exactly, and its packed covariance coordinate is $\widehat{X}_n/\sqrt{2}$.

Here are fully explicit bounds behind the root. On $\mathcal{E}_n = \{|S - d_n| \leq 2\sqrt{d_n},\ |T| \leq 1\}$, Chebyshev's inequality, independence, and $\mathbb{P}(|T| \leq 1) > 1/3$ give $\mathbb{P}(\mathcal{E}_n) > 1/6$. On this event $|s_\delta| \leq 3$. Since Euler's number satisfies $\mathrm{e} < 3$, uniformly for $0 \leq \delta \leq 1$,

$$\frac{1}{4704\sigma_n} \leq Q_{d_n,\sigma_n}(\delta) \leq \frac{1}{4\sigma_n}, \qquad \frac{1}{672R_n} \leq M_{d_n,\sigma_n}(\delta) \leq \frac{3}{R_n}. \tag{7.3}$$

For the lower bounds use $p \geq 1/28$, $q \geq 1/(784\sigma_n)$, and $\operatorname{Tr} D \geq d_n/(4R_n)$ on $\mathcal{E}_n$. For the upper bound on M, use

$$\frac{1}{d_n} p\left(\frac{d_n - 1}{r} + \frac{\sigma_n^2}{r^3}\right) \leq \frac{1}{r}, \qquad \frac{1}{4\sigma_n d_n} \leq \frac{1}{R_n}, \qquad \mathbb{E}r^{-1} \leq (\mathbb{E}S^{-1})^{1/2} = (d_n - 2)^{-1/2} \leq 2R_n^{-1}.$$

Since $R_n/\sigma_n = \sqrt{n^2+1} > 14112$, the left side minus the right side of (3.5) is negative at zero and positive at one. Thus the specified root exists, and (7.3) gives

$$\frac{\sigma_n}{168R_n} \leq \delta_n^2 \leq 14112\frac{\sigma_n}{R_n}, \qquad \frac{1}{672R_n} \leq m_n \leq \frac{3}{R_n}, \qquad \frac{R_n}{6} \leq c_n \leq 672R_n. \tag{7.4}$$

In particular, the two equalities in (3.11) remain exact; the inequalities in this paragraph are used only to audit scales.

Define the explicit valid upper curvature and class ceiling

$$\widetilde{\beta}_n := 1009\frac{R_n}{\sigma_n}, \qquad \overline{\kappa}_n := \frac{\widetilde{\beta}_n}{\alpha_n} = 1009(n^2+1). \tag{7.5}$$

Equations (2.6) and (7.4) give $\alpha_n I \preceq W_n \preceq \widetilde{\beta}_n I$. Moreover the direct chain-rule estimate (4.7) can be read with constant eight, so

$$L_H(V_n) \leq \frac{8c_n}{\sigma_n^2} \leq 5376\frac{R_n}{\sigma_n^2} \leq 8064 =: \Lambda_0. \tag{7.6}$$

Consequently $(N_\theta^{(n)}, V_n) \in \mathcal{C}(\overline{\kappa}_n, \Lambda_0)$ and $\overline{\kappa}_n \to \infty$.

For completeness, the constants in the normalized Rayleigh calculation can also be made numerical. Equations (7.4), (5.7), and (5.9) give

$$d_n\lambda_n^2 \le 112896\alpha_n, \qquad \mathbb{E}[(2,\lambda_n Y)^\top A_n(2,\lambda_n Y)] \le \frac{\lambda_n^2}{2\sigma_n} + 2\frac{\lambda_n^2\sigma_n^2}{R_n}.$$

Using $R_n^2/d_n \le 2$, $R_n^2/d_n^2 \le \alpha_n$, and $\sigma_n^3 R_n/d_n^2 \le \alpha_n$, the two terms after multiplication by c_n are at most $18966528\alpha_n$ and $75866112\alpha_n$, respectively. Thus

$$c_n\mathbb{E}[(2,\lambda_n Y)^\top A_n(2,\lambda_n Y)] \le 94832640\alpha_n.$$

Together with the floor energy and the final $\|X\|_F^2$ term in (5.2), which are at most $112900\alpha_n$ and $112896\alpha_n$, respectively, this yields

$$4\mathcal{Q}_{W_n}(e_t,\lambda_n P_{d_n}) < 200000000\alpha_n.$$

For the literally normalized witness (5.5), therefore,

$$\gamma_{\mathrm{loc}}(V_n) \le 50000000\alpha_n \le \frac{1600000000}{\sqrt{\overline{\kappa}_n}}. \tag{7.7}$$

If κ_n^{opt} denotes the optimal post-whitening condition number, the point $y=0$, $s=0$ and (7.5) give

$$\frac{n^2+1}{12} \le \kappa_n^{\mathrm{opt}} \le 1009(n^2+1). \tag{7.8}$$

The same last bound in (7.7), with the same numerical constant, therefore holds with κ_n^{opt} in place of $\overline{\kappa}_n$. Hence outcome (B) holds with

$$\psi(\kappa) := 1600000000\,\kappa^{-1/2}, \qquad \psi(\kappa)(1+\log\kappa) \longrightarrow 0. \tag{7.9}$$

8 Final guardrail audit

The construction requires no coordinate transfer: it is defined with $Z \sim N(0,I)$ and satisfies $\mathbb{E}W(Z) = I$ exactly. Thus every $\alpha, \beta, \kappa, L_H$ above is already post-whitening. The remaining guardrails are met as follows.

1. Equation (2.5) is the pointwise Hessian of the displayed C^∞ potential. Full third- and fourth-order compatibility, not first-Hermite symmetry alone, is used in Lemma 5.1 and (6.2).
2. No commutation-free Loewner transfer occurs. The inverse gap is the scalar Rayleigh calculation (5.7)–(5.14).
3. Bounds and isotropic mean are not treated as sufficient abstract data. Exact isotropy comes from (3.5), and the small energy comes from the compatible radial and mixed blocks of (2.5).
4. Both κ and L_H are post-whitening quantities; no raw-coordinate estimate is imported.
5. The Hessian and every derivative of the Hessian are bounded, so every Gaussian integration by parts used here is classical. No unproved weak extension is needed.
6. The inverse gap is $\Theta(\sqrt{\kappa})$, genuinely larger than $O(\log\kappa)$, even when $L_H \to 0$.
7. No statement about a nonlinear attraction radius is made.
8. At no point is an operator bound on $\nabla^3 V$ converted into a Frobenius bound. The large Frobenius coupling is instead certified directly by the normalized test (6.1).